

MILITARY MISSILE

The US Navy tests an early surface-to-air missile (SAM) in 1950. SAMs were to revolutionize air defenses and place new demands on aircraft design.



their independence from the United States, produced high-performance fighter aircraft such as the Super Mystère and the Mirage III through the Dassault company in the 1950s, and went on to develop an independent nuclear strike force in the following decade. Britain was also committed to developing its own nuclear deterrent in the 1950s and produced effective aircraft for offense and defense: the three “V bombers” – Victor, Valiant, and Vulcan – and the Hawker Hunter and English Electric Lightning interceptors. But by the time the Lightning entered service in 1959, economic stringencies were forcing Britain to drop out of independent development of aircraft for strategic nuclear war.

Missile development

One of the reasons the British government gave for canceling expensive fighter-interceptor projects was that air defense would soon be exclusively a business for ground-based missiles. The increasing effectiveness of surface-to-air missiles (SAMs) in the late 1950s led to some major rethinking all around on strategic air war. The nuclear bomber forces were geared up to

penetrate the enemy’s defenses through speed and high altitude. But once SAMs had proved they could shoot down high-flying aircraft, the bombers had to turn to very low-level attack, attempting to fly under enemy radar and missile screens.

“I was a pilot flying an airplane and... where I was flying made what I was doing spying.”

FRANCIS GARY POWERS

INTERVIEWED AFTER HIS RETURN TO THE US

Cold War spyplanes

That Soviet SAMs could strike aircraft at high altitude was proven in 1960 when a U-2 spyplane

was downed over Sverdlovsk. The U-2 was designed to operate at an altitude of more than 70,000ft (21,000m), which was hoped to be out of reach of Soviet radar, ground-based missiles, and interceptors. Run by the CIA, U-2s overflew the Soviet Union repeatedly from 1956, bringing back detailed photographs of military installations. In May 1960, however, a U-2 piloted by Francis Gary Powers was sent into a fatal spin by a SAM exploding nearby. Powers escaped from the cockpit and parachuted to the ground, having failed to activate the U-2’s self-destruct mechanism, and declined to use the cyanide pill provided to avoid capture. Soviet leader Nikita Khrushchev drew maximum diplomatic advantage from the incident, flouncing out of a

AIR-TO-AIR MISSILES

THE FIRST GENERATION of air-to-air guided missiles (AAMs) was developed in the 1940s. They were either radar controlled or heat-seekers. The radar-controlled missiles, such as the Falcon and Sparrow, were designed for intercepting high-altitude bombers. A radar on board the interceptor locked on to its target and the missile homed in on the reflections from the interceptor’s radar. Keeping the target illuminated with the radar during the missile’s flight meant that the interceptor had to stay both straight and level –

meaning that the missile was useless for combat between fighters.

In contrast, the heat-seeking Sidewinder was a fire-and-forget missile. In classic dogfight style, the

**FALCON MISSILES**

A pilot shows off radar-guided (left) and heat-seeking Falcon missiles, which came into service in the mid-1950s.

**RADAR-GUIDED SPARROW**

A US Navy F-14A Tomcat fires an AIM-7 Sparrow radar-guided, air-to-air missile. The Sparrow, which is widely deployed by US and NATO forces, can destroy targets more than 10 miles (16km) away.

pilot maneuvered his aircraft behind the enemy’s tail and fired the missile toward the hot exhaust. An infrared detector in the missile guided it on to the heat source. Although effective, until the 1970s heat-seeking missiles had drawbacks: they were useless fired head-on and could easily pick up the wrong target if another jet exhaust presented itself.

Inevitably, guided missiles of all kinds bred countermeasures. Radar-guidance systems could be jammed or confused by chaff; heat-seekers could be distracted by firing flares or shaken off by swift maneuvering.